

Shirui Zhou

PhD Candidate

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Profile

Shirui Zhou is a researcher working at the intersection of complex systems, human mobility, and sustainable transportation systems. His research combines statistical physics, computational modelling, and data-driven approaches to investigate collective mobility dynamics, behavioural adaptation, traffic instability, and mobility-energy interactions. His recent work increasingly explores accessibility inequality, sustainable urban systems, and AI-enhanced modelling for large-scale socio-technical systems.

Research Interests

- **Human Mobility and Collective Dynamics:** Traffic dynamics, collective mobility behaviour, behavioural adaptation, and multi-scale transportation systems.
- **Complex Systems and Computational Modelling:** Statistical physics, nonlinear dynamics, stochastic systems, and data-driven modelling of socio-technical systems.
- **Sustainable Mobility and Urban Systems:** Accessibility inequality, electric mobility, mobility-energy interactions, and sustainable urban systems.

Education

Ph.D. Candidate in Management Science and Engineering Institute of Systems Engineering, Tianjin University, Tianjin, China MATS Laboratory	Sep. 2023–Jun. 2027
Visiting Ph.D. in Complexity Physics Department of Physics, Sapienza University of Rome, Rome, Italy Sony Computer Science Laboratories, Rome, Italy Host: Prof. Vittorio Loreto	Aug. 2025–Aug. 2026
M.Eng. in Systems Engineering Institute of Systems Engineering, Tianjin University, Tianjin, China	Sep. 2021–Dec. 2023
B.E. in Transportation; B.S. in Statistics Southwest Jiaotong University, Chengdu, China	Sep. 2017–Jun. 2021

Papers

Published

1. A Driving Regime-Embedded Deep Learning Framework for Modeling Intradriver Heterogeneity in Multiscale Car-Following Dynamics. *IEEE Transactions on Cybernetics*. arXiv:2506.05902. **First author.**
2. Cellular Automaton Model With the Multi-Anticipative Effect To Reproduce the Empirical Findings of Kerner's Three-Phase Traffic Theory. *Physica A*. **First author.**
3. Cellular Automaton Approach for Modeling Traffic System. *Springer*. **Co-author.**
4. On the Calibration of Stochastic Car Following Models. *Transportation Research Part B*. arXiv:2302.04648. **First author.**
5. Experimental Features of Emissions and Fuel Consumption In a Car-Following Platoon. *Transportation Research Part D*. arXiv:2305.00750. **First author.**
6. Towards Experiment-Driven and AI-Enhanced Driving Behavioral Research. *Engineering Management*. **First author.**

arXiv Preprints

1. A Structured Framework for Calibrating Stochastic Car-Following Models: Data Adequacy, Parameter Sensitivity, and Objective Selection. *arXiv preprint*. arXiv:2608.18944. **First author.**

2. The Moving Target of Urban Equity: Spatiotemporal Demand and Double Disadvantage In Hefei, China. *arXiv preprint*. arXiv:2606.20132. **First author**.
3. Human adaptive variability stabilises collective traffic dynamics. *arXiv preprint*. arXiv:2606.26868. **First author**.
4. Twenty-Five Years of the Intelligent Driver Model: Foundations, Extensions, Applications, and Future Directions. *arXiv preprint*. arXiv:2506.05909. **First author**.
5. A Strictly Proper Scoring-Rule Theory for Calibrating Stochastic Car-Following Models. *arXiv preprint*. arXiv:2609.04988. **First author**.

Research Projects

National Natural Science Foundation of China Key Project	Jan. 2025 – Dec. 2029
• Human-machine integration systems: theory, operation, and risk management.	
National Natural Science Foundation of China Excellent Young Scholars Program	Jan. 2023 – Dec. 2025
• Urban traffic behavior analysis and management.	
National Natural Science Foundation of China International Cooperation Project	Oct. 2023 – Oct. 2027
• Smart management and resilience enhancement of urban road traffic systems in connected and autonomous driving environments.	
Beijing-Tianjin-Hebei Basic Research Cooperation Project	Sep. 2024 – Sep. 2026
• Theory and technology of intelligent traffic control under the collaborative environment of people-vehicle-road-cloud systems.	